

1 Estimate PID's death time



GC read



Buffer Manager

IO Interface

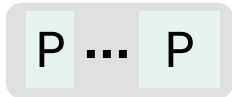
2 Compress PIDs



3 Group them by EDT and page pack PIDs per group

group 1

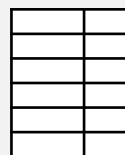
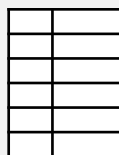
group 2



GC write

Space Manager

PID2Offset Offset2PIDs



ActiveGroupHistory

G1(Z0,Z1)
G2(Z2,Z3)
⋮

StorageSpace

empty

active

full

z

z

z

z

z

⋮

⋮

⋮

⋮

⋮

z

z

z

z

z

aysnc

GC



4 Select a zone per group

group 1: Z0

group 2: Z19

GC valid PIDs

5 If there is free or active zone available,

① Return zone ID with closest deathtime

② Otherwise, select zone from the full list to GC

→ Select victim zones depending on frequency imbalance among active groups

Z0

Z19

...

SSD

6 Issue writes per group